ORIUS

Observation–Reality Integrity for Universal Safety

A Universal Safety Monograph for Physical AI

Runtime Safety Under Degraded Observation

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A universal runtime safety layer developed across battery, autonomous vehicle, industrial, healthcare, navigation, and aerospace domains.

ORIUS Safety Principle

"Physical AI safety is bounded by observation reliability."

April 2026

Abstract

Physical AI systems do not act on reality directly. They act on measurements, filtered state estimates, delayed telemetry, and learned summaries of the world. When that observation surface degrades, an action can remain admissible on the observed state while already being unsafe on the true state. This dissertation treats that discrepancy as an action-semantics safety problem rather than as a prediction problem alone, and names it the observation–action safety gap (OASG).

Existing lines of work address important parts of this failure mode: conformal uncertainty quantifies predictive spread, safe control constrains actions, runtime assurance supervises release, and fault monitoring detects degraded telemetry. What is still missing is the integrated runtime object that connects those pieces at the moment of physical release. ORIUS, Observation–Reality Integrity for Universal Safety, is proposed as that missing object: a certificate-aware runtime safety layer for physical AI under degraded observation.

The dissertation develops ORIUS through three implementation-grounded surfaces. First, `DomainAdapter` makes the domain safety predicate, fallback semantics, and repair geometry explicit. Second, a universal benchmark schema turns degraded-observation replay into a common evaluation contract across domains. Third, CertOS provides certificate lifecycle, audit continuity, and release semantics so that repaired action can be governed as a runtime object rather than reported only as controller-side logic. The formal contribution is a domain-agnostic theorem ladder connecting degraded observation, reliability-conditioned uncertainty, repaired action, fallback, and certificate semantics under explicit assumptions.

Empirically, the dissertation establishes one deep witness row and several bounded multi-domain instantiations without changing what ORIUS means at the release boundary. Battery provides the strongest theorem-to-runtime closure, and industrial process control plus healthcare monitoring support defended bounded rows. The result is a thesis that is universal in architecture, tiered in evidence: the autonomous-vehicle row is a defended bounded runtime row under the current artifact surface, while canonical raw-data closure remains incomplete, and equal-domain closure remains open because navigation and aerospace are still gated.

Acknowledgments

This thesis was assembled as a solo research project around one question: how should a physical-AI system behave when the observation channel is no longer trustworthy, but the consequences of action remain physical and irreversible? ORIUS was built to answer that question with a runtime contract rather than a domain-specific controller.

The deepest current evidence still comes from the battery witness row because it offers the cleanest theorem-to-code-to-artifact surface in the repository. The broader contribution of the book, however, depends on carrying that safety grammar across autonomous vehicles, industrial control, healthcare monitoring, navigation, and aerospace without letting the prose outrun the evidence.

The open-source systems, datasets, and research communities behind the methods cited in this book made that synthesis possible. Their work is visible throughout the bibliography, the artifact index, and the reviewer-facing appendices that helped harden the final manuscript.

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Part I

Problem, Principle, and Gap

Chapter 1

Introduction and Thesis Claims

A battery controller can be shown stale feeder telemetry and continue dispatching as if the network were quiet. A vehicle can track a nominal lane geometry after the observation stream has already fallen behind the scene. A bedside monitoring system can smooth away a sudden physiological transition while its downstream policy still appears internally consistent. In each case the failure is structurally the same: the controller acts on an observation surface that is coherent enough for computation but no longer trustworthy enough for safe release. This dissertation begins from that release-boundary hazard.

1.1 From degraded observation to unsafe release

Physical AI systems always act through an observation layer. Telemetry arrives late, state estimators compress uncertainty, learned models summarize partially observed trajectories, and communication channels drop, reorder, or distort critical signals. Most stacks treat those effects as nuisances to be absorbed by better prediction, better perception, or better control tuning. ORIUS starts from the opposite premise: once degraded observation changes what can be known about the physical world, it changes what actions are safe to release.

The central danger is therefore not forecast error alone and not perception error alone. It is the possibility that action legality on the observed state diverges from action legality on the true state. ORIUS names that discrepancy the *observation–action safety gap* (OASG). The OASG is the dissertation's scientific object because it identifies the exact place where nominal intelligence can remain coherent while physical safety has already been lost.

1.2 Thesis claim

This dissertation advances one central claim: *physical AI under degraded observation requires a runtime safety layer that sits between nominal intelligence and physical release, and that layer can be expressed through one reusable grammar of detection, uncertainty inflation, safe-set tightening, action repair or fallback, and certificate-aware release.* ORIUS is proposed as that layer.

The claim is strong but intentionally bounded. ORIUS is not a universal optimal controller, not a replacement for domain-specific control design, and not yet a proof of equal-domain closure across every supported application. It is a runtime discipline for deciding what action may still be released once observation quality has degraded.

1.3 ORIUS as the dissertation object

The reusable object in this dissertation is a typed Detect–Calibrate–Constrain–Shield–Certify kernel.

- Detect scores reliability loss in the observation channel.
- Calibrate widens the observation-consistent state set.
- Constrain recomputes admissible action under that widened set.
- Shield repairs or replaces unsafe candidate actions.
- Certify records why the release remains admissible and how long that claim should remain valid.

The important point is that ORIUS does not compete with every controller on the same terms. It addresses the release-boundary decision that appears once a controller admits the observation channel may already be wrong.

Three implementation-grounded surfaces make this claim more than conceptual language. `DomainAdapter` exposes the domain safety predicate, true-state margin, and fallback semantics that the runtime must protect. The universal benchmark schema exposes one replay contract for true-state violation, intervention, fallback, certificate validity, and latency. CertOS supplies the lifecycle semantics that determine when repaired action can be issued, renewed, expired, revoked, or replaced by fallback.

Those three surfaces are not supporting details around a more abstract contribution. They are the mechanism that makes ORIUS scientifically concrete. Without `DomainAdapter`,

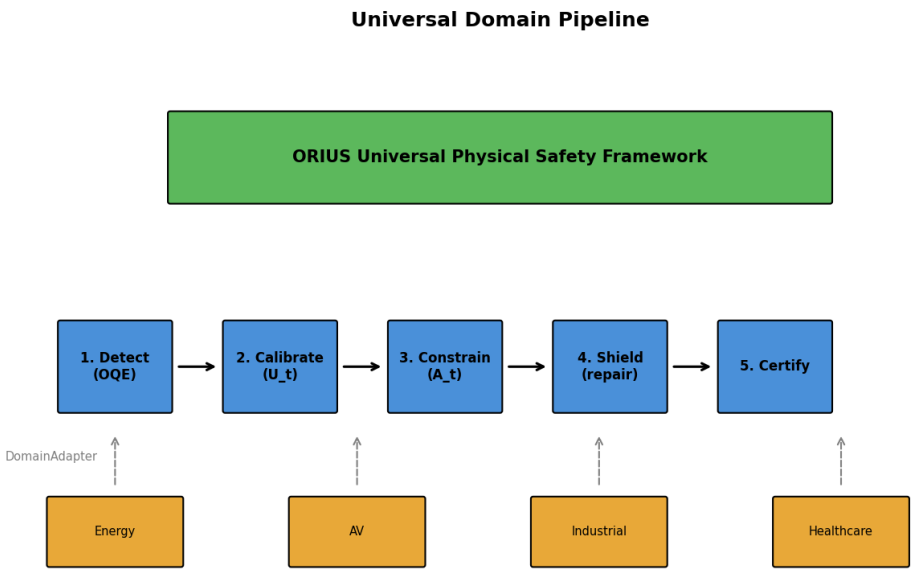

Figure 1.1: ORIUS as a universal runtime safety layer between nominal intelligence and physical release. The same five-stage kernel recurs across domains even though the domains do not share a plant model, controller family, or optimization objective.

the safety object stays implicit and domain-local. Without the universal benchmark schema, cross-domain degraded-observation claims remain incomparable. Without CertOS, repaired action lacks a governed release semantics. Taken together, they are what turns ORIUS from a collection of useful ideas into a reusable runtime layer.

1.4 Main contributions

The dissertation makes exactly four contributions.

1. **A runtime object.** It defines OASG-centered safety under degraded observation as a first-class runtime problem for physical AI.

2. **A systems object.** It presents the ORIUS kernel as an adapterized runtime safety layer built around `DomainAdapter`, the universal benchmark schema, and CertOS release semantics.

3. **A formal object.** It develops a theorem bridge from reliability-conditioned uncertainty and repaired action to bounded violation reduction under explicit assumptions.

4. **An empirical object.** It assembles bounded multi-domain evidence with one witness row and explicit defended and gated rows rather than flattening all domains into a false claim of equal maturity.

1.5 Claim discipline

The manuscript uses three claim classes and keeps them distinct.

Proved claims theorem-level results under stated assumptions for OASG, repair, and bounded safety improvement.

Validated claims replay-backed and runtime-governed empirical claims tied to tracked artifacts.

Roadmap claims extensions that are structurally motivated but not promoted as defended evidence.

That distinction keeps the dissertation from confusing architectural universality with empirical parity. The exact row-by-row maturity boundary is stated in the cross-domain chapter and then restated once in the conclusion, rather than repeated throughout the manuscript.

1.6 Why ORIUS matters beyond one domain

The importance of ORIUS is not that batteries, vehicles, industrial plants, healthcare systems, navigation stacks, and aerospace controllers secretly reduce to the same plant. The importance is that they all face the same release-boundary question once observation trust degrades: what action can still be justified, with what margin, and under what fallback and certificate? ORIUS freezes the semantics of that question so that cross-domain comparison happens at the level of runtime safety rather than only at the level of application area.

1.7 Dissertation organization

The rest of the dissertation follows that argument directly. Chapters 2 and 3 define the novelty gap and formal problem object. Chapters 4 through 6 present the ORIUS kernel, theorem ladder, and replay protocol. Chapters 7 and 8 defend the battery witness row. Chapters 9 through 12 test how far the same runtime grammar carries across domains and governance surfaces. Chapter 13 then states the precise evidence boundary and the final thesis claim.

Chapter 2

Related Work and the Novelty Gap

2.1 Conformal prediction and adaptive calibration

Conformal prediction and related distribution-free uncertainty methods provide one of the few disciplined ways to wrap learned models with coverage-aware uncertainty without assuming a perfectly specified error law [1–6]. Conformalized quantile regression and adaptive conformal variants extend that logic to heteroskedastic and nonstationary settings, which makes them especially relevant once a physical system moves through regime changes or degraded telemetry states [3, 4, 7–11].

Why this is insufficient for ORIUS. This family solves uncertainty quantification; it does not solve the release-boundary question that follows from that uncertainty. Coverage guarantees do not by themselves decide how the admissible action set should change, which fallback is allowed, or what evidence must accompany release once the observation surface is unreliable. ORIUS therefore uses calibration as an input to runtime safety, not as the full safety object.

2.2 Runtime assurance and Simplex-style supervision

Runtime assurance architectures, supervisory controllers, shielding, and runtime verification supply the language of intervention, veto, fallback, and monitored execution [12–20]. That line of work is indispensable because it treats safe release as a separate problem from nominal policy design.

Why this is insufficient for ORIUS. Most supervisory schemes enter after the nominal controller has already proposed an action under a state estimate that is treated as sufficiently

meaningful. ORIUS addresses an earlier fault line: once observation quality degrades, the assurance layer must reason jointly about reliability, widened state uncertainty, repaired action, fallback, and certificate semantics. A supervisor without that observation-side logic is still missing the central OASG object.

2.3 Safety filters, shields, and robust safe control

Control barrier functions, predictive safety filters, robust MPC, and tube-based control translate model uncertainty into admissible control sets and corrective intervention rules [21–31]. These methods are the closest classical relatives of ORIUS because they formalize the difference between nominal control intent and safe control execution.

Why this is insufficient for ORIUS. This family usually assumes that the uncertainty set and safety object have already been defined in a domain-specific way. ORIUS does not replace that work with a universal controller. Instead, it provides a typed runtime contract that can host many repair mechanisms while keeping reliability estimation, fallback semantics, and certificate discipline explicit. The novelty is not the existence of safe control alone; it is the runtime composition that makes safe control portable across domains.

2.4 Anomaly, drift, and degraded-observation detection

Fault diagnosis, change detection, cyber-physical security, and anomaly monitoring provide the vocabulary for stale, delayed, reordered, dropped, or corrupted observations [32–42]. This literature is essential because degraded observation must first be recognized before it can influence action release.

Why this is insufficient for ORIUS. Detection alone does not specify how action semantics should change after trust degrades. A drift alarm can tell the system that something is wrong without saying how uncertainty inflates, how the safe set tightens, what repair is permitted, or what certificate makes the next step auditable. ORIUS uses detection as the opening move in a runtime safety sequence, not as the full response.

2.5 Domain-specific cyber-physical safety systems

Energy systems, autonomous mobility stacks, industrial control, healthcare monitoring, and other cyber-physical systems already contain rich safety practice in their own local forms

[12, 21, 32, 34–36]. This practice matters because any claim of universality that ignores domain structure is empty from the start.

Why this is insufficient for ORIUS. Domain-specific systems typically optimize within one application vocabulary and one local evidence regime. They rarely expose a common contract that allows degraded-observation replay, repaired action, fallback semantics, certificate validity, and latency to be compared across domains. ORIUS addresses that missing cross-domain runtime object rather than trying to erase domain structure.

2.6 The novelty gap ORIUS addresses

The literature becomes most powerful when these families are read together, but it also reveals a persistent separation of levels. Uncertainty methods stop at calibrated sets. Assurance methods stop at supervisory intervention. Safe-control methods stop at plant-specific repair geometry. Detection methods stop at identifying degraded telemetry. Domain systems stop at application-local safety practice. What is missing between them is a single typed runtime composition that starts with observation reliability and ends with repaired action, fallback, certificate semantics, and cross-domain replay under one common contract.

That missing composition is ORIUS's novelty claim. The dissertation does not argue that any one ORIUS component is unprecedented in isolation. It argues that the combination of observation reliability, action repair, certificate semantics, and universal replay has not previously been stabilized as one reusable runtime safety layer for physical AI. ORIUS is therefore not "better forecasting," "better perception," or "one more controller." It is a release-boundary discipline for degraded-observation safety.

The closest plausible rival to ORIUS is not any single paper in these families. It is an imagined composition of them: adaptive conformal intervals for uncertainty, a safety filter or Simplex supervisor for intervention, anomaly detection for degraded telemetry, and runtime verification for logging and policy enforcement. Even that composite stack still stops short of ORIUS's central object. It lacks a typed release contract that binds observed-state and true-state semantics, repaired action, fallback, certificate lifecycle, and cross-domain replay under one explicit interface. That final integration step is the novelty claim the dissertation defends.

Chapter 3

Formal Problem Formulation: OASG under Degraded Observation

3.1 State, observation, action, and constraints

Let $x_t^\star$ denote the true physical state, $\hat{x}_t$ the observation available to the nominal controller, and u_t the candidate action proposed at time t. Let $\mathcal{C}(x)$ denote the safety constraints that define admissible actuation for a given state. ORIUS is interested in regimes where the observation surface becomes a poor enough proxy for the true state that a nominally legal action can become physically unsafe.

3.2 Observation degradation operator

Observation degradation is modeled as a runtime operator that maps truthful telemetry into an imperfect observation stream:

$$\hat{x}_t = \mathcal{D}(x_t^\star; \delta_t),$$

where δ_t summarizes stale packets, packet loss, delay, reordering, spikes, drift, or other telemetry faults. The universal object in the dissertation is not a plant model alone; it is the pair $(x_t^\star, \hat{x}_t)$ together with the fault process that causes them to diverge.

3.3 Definition of the observation–action safety gap

The central event is the observation–action safety gap:

$$u_t \in \mathcal{C}(\hat{x}_t) \qquad \text{but} \qquad u_t \notin \mathcal{C}(x_t^\star).$$

In words, an action is safe on the observed state while unsafe on the true state. This is the formal object that later chapters measure through true-state violation metrics, repair semantics, and certificate validity.

3.4 Observation-consistent state sets

ORIUS does not try to recover the true state exactly at runtime. Instead, it constructs an observation-consistent state set

$$\mathcal{X}_t = \{x : x \text{ remains plausible given } \hat{x}_t \text{ and reliability } w_t\},$$

where $w_t \in [0, 1]$ is the runtime reliability score. As reliability decreases, $\mathcal{X}_t$ widens, and the admissible action set must tighten accordingly.

3.5 Dual-trajectory evaluation

This dissertation evaluates ORIUS on *dual trajectories*. The observed trajectory is the one seen by the nominal controller. The true trajectory is the one used to determine whether the repaired or unrepaired action actually violates the physical envelope. A benchmark that only inspects observed-state legality cannot see OASG directly. ORIUS-Bench therefore records candidate action, repaired action, true-state violation, observed-state satisfaction, fallback/intervention markers, latency, and certificate validity in one replay schema.

3.6 DomainAdapter grounding

The universal formulation is grounded in the repository through the domain adapter contract. Each domain supplies:

1. telemetry parsing and domain state semantics,
2. uncertainty or margin construction hooks,
3. admissible-action or repair logic,
4. true-state violation predicates,
5. observed-state satisfaction predicates, and
6. certificate payload specifics.

That boundary lets the dissertation make a strong universal claim about the runtime grammar without pretending that one plant model, one objective, or one controller class solves every domain.

3.7 Scope and assumptions

The problem formulation is intentionally narrower than a claim of general AI safety. ORIUS assumes an actionable physical system with explicit safety constraints, an observation channel that can degrade, and a runtime layer that can intercept or repair actions before release. It does not assume perfect state recovery, universal optimal control, or equal empirical maturity across all domains. Those boundaries are what make the later theorem and validation claims durable rather than aspirational.

Part II

ORIUS Framework and Theorem Ladder

Chapter 4

ORIUS Architecture: DC3S Runtime Layer

ORIUS is implemented as a post-nominal runtime layer. It does not replace the upstream forecaster, planner, or controller. It wraps them with a typed safety contract that translates degraded observation into repaired action and certificate-aware governance.

4.1 Why a runtime layer is needed

The dissertation does not assume that every legacy controller can be redesigned around one global proof technique. What can be reused is the safety layer that mediates between nominal intelligence and actuation. That is the architectural choice that makes ORIUS cross-domain: domain-specific intelligence remains domain-specific, while degraded observation, action tightening, fallback, and audit semantics are shared.

4.2 The Detect–Calibrate–Constrain–Shield–Certify step

The universal runtime step consists of five stages.

Detect compute observation reliability from freshness, missingness, delay, ordering, spikes, and consistency.

Calibrate widen the observation-consistent state set as reliability drops.

Constrain recompute the admissible action set under the widened uncertainty set.

Shield repair the candidate action or replace it with a fallback when no safe action remains.

Certify emit a causal record describing reliability, uncertainty, repair result, and lifecycle status.

This step is deterministic at the contract level even when the nominal controller is learned, stochastic, or domain-specific.

4.3 Universal runtime flow

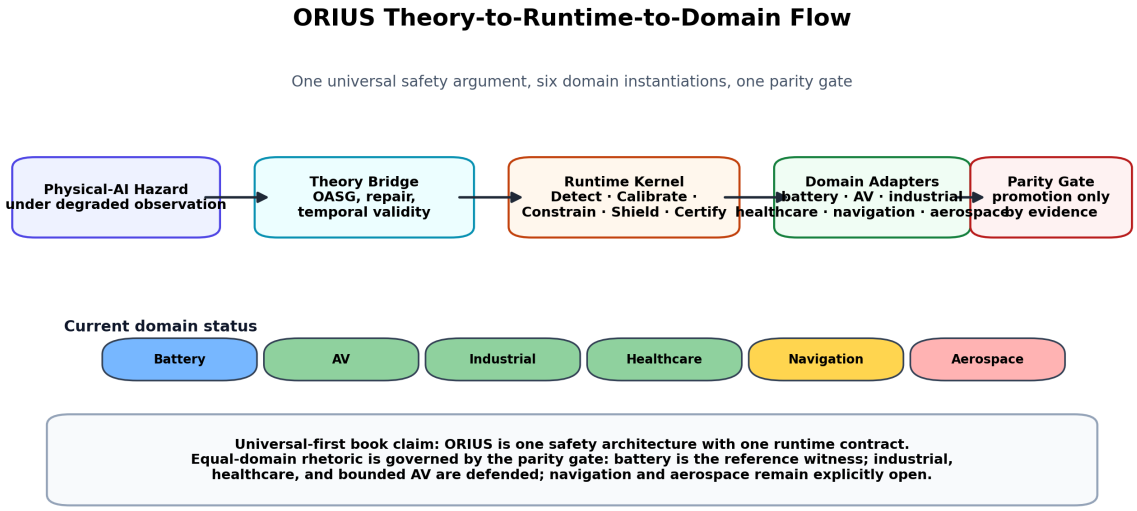

Figure 4.1: ORIUS runtime flow from degraded telemetry to repaired action and governed certificate. The figure is generated from the same publication assets that feed the parity-facing manuscript surfaces.

4.4 DomainAdapter contract

The adapter contract is the narrow universal boundary between plant-specific semantics and shared runtime logic. The adapter supplies state interpretation, safety predicates, repair/fallback hooks, and certificate details. The kernel supplies reliability handling, uncertainty inflation, replay schema compatibility, and lifecycle discipline. This is the key design choice behind the dissertation's universality claim: one runtime grammar, many domain instantiations.

Table 4.1: Domain state schema for ORIUS universal framework.

Domain	State Fields	Unit	Des
Energy	soc_mwh, load_mw, renewables_mw	MWh, MW	Bat
AV	position_m, speed_mps, speed_limit_mps, lead_position_m	m, m/s	Lon
Navigation	x, y, vx, vy	m, m/s	2D
Industrial	temp_c, pressure_mbar, humidity_pct, power_mw	°C, mbar, %, MW	Pro
Healthcare	hr_bpm, spo2_pct, respiratory_rate	bpm, %, /min	Vita
Aerospace	altitude_m, airspeed_kt, bank_angle_deg, fuel_remaining_pct	m, kt, deg, %	Flig

Table 4.2: Safety predicates per domain.

Domain	Safety Predicate	Config Key
Energy	SOC $\in$ [min_soc, max_soc]; charge/discharge mutual exclusion	min_soc_mwh
AV	$v \leq v_{\max}$; $x_{\text{lead}} - x \geq d_{\min}(v)$	speed_limit_mps
Navigation	(x, y) remains inside arena bounds and outside obstacle set	arena_min, arena_max
Industrial	temp $\in [\mathrm{T}_{\min}, \mathrm{T}_{\max}]$; power $\in [0, \mathrm{P}_{\max}]$	power_max_mw
Healthcare	hr $\in [\mathrm{hr}_{\min}, \mathrm{hr}_{\max}]$; $\mathrm{SpO}_2 \geq \mathrm{spo2}_{\min}$	spo2_min_pct
Aerospace	$v \in [\mathrm{v}_{\min}, \mathrm{v}_{\max}]$; $\lvert\mathrm{bank}\rvert \leq \mathrm{bank}_{\max}$; fuel $\geq \mathrm{fuel}_{\min}$	v_min_kt, max_bank_

4.5 Runtime cost and operational realism

ORIUS is written for operational deployment rather than only for theorem exposition. Runtime overhead, fallback behavior, and auditable certificates are not appendix-only concerns. They are part of the scientific object because a safety layer that cannot act in time, cannot explain its interventions, or cannot preserve governance continuity is not a defensible physical-AI runtime layer.

4.6 What the architecture does and does not claim

The architecture claims a reusable runtime contract. It does not claim universal optimal repair, universal plant realism, or certification compliance. Those stronger statements depend on the domain row, benchmark replay, and governed artifact surface that later chapters make explicit.

Table 4.3: Universal DC3S pipeline stages.

Stage	**Name**	**Adapter Method**	**Output**
1	Detect	compute_oqe	w_t, flags
2	Calibrate	build_uncertainty_set	$\mathcal{U}_t$, meta
3	Constrain	tighten_action_set	$\mathcal{A}_t$
4	Shield	repair_action	safe_action, repair_meta
5	Certify	emit_certificate	certificate

Chapter 5

Mathematical Foundations and Safety Guarantees

5.1 Assumptions and theorem discipline

The dissertation keeps the theorem surface intentionally explicit. Every strong guarantee depends on a named assumption set, and those assumptions are recorded in the assumption register in Appendix a. The point of the theory chapter is not to overclaim generality. It is to show exactly which parts of the ORIUS argument are proved, which are validated empirically, and which remain bounded by current evidence.

5.2 The theorem ladder

The current ORIUS theorem ladder establishes four core ideas:

1. OASG can arise even when observed-state legality appears intact.
2. Reliability-aware repair preserves safety under stated uncertainty and feasibility assumptions.
3. The risk envelope tightens as reliability degrades, so intervention pressure is a lawful consequence of degraded observation rather than an arbitrary heuristic.
4. There is no free safety: quality-ignorant control can be driven into true-state violation by admissible degradation regimes.

The deepest theorem-to-code-to-artifact chain remains battery-backed, but the theorem objects themselves are written against universal typed contracts and runtime objects.

Table 5.1: DC3S Theorem Dependency and Proof Completeness. Each formal result is implemented in code and tested by at least one automated test.

Result	Name	Depends On	Proof Method	Code Referen
Defn. 1	OQE reliability weight w_t	—	Definition	`quality.py:co`
Prop. 1	Inflation monotonicity	Defn. 1	Analytic	`coverage_theo`
Thm. 1	Marginal coverage guarantee	Prop. 1, Defn. 1	Conformal CP theory	`coverage_theo`
Thm. 3	Expected violation bound	Thm. 1	Expectation calculus	`coverage_theo`
Thm. 9	Group-conditional coverage	Thm. 1	Mondrian CP	`coverage_theo`
Thm. 10	Hoeffding high-prob. bound	Thm. 3	Hoeffding inequality	`coverage_theo`
Thm. 11	Byzantine OQE reliability bound	Defn. 1	Trimmed-mean theory	`quality.py:co`

All results are unit-tested in `tests/test_conditional_coverage.py` and `tests/test_adversarial_faults.p`

5.3 Risk envelope interpretation

ORIUS does not prove safety from a single forecast accuracy number. The risk envelope is expressed in terms of uncertainty coverage, miscoverage, and mean reliability. That is why the dissertation can connect theorem statements to runtime calibration and certificate semantics without collapsing them into a purely statistical paper or a purely control paper.

5.4 Theorem-to-artifact linkage

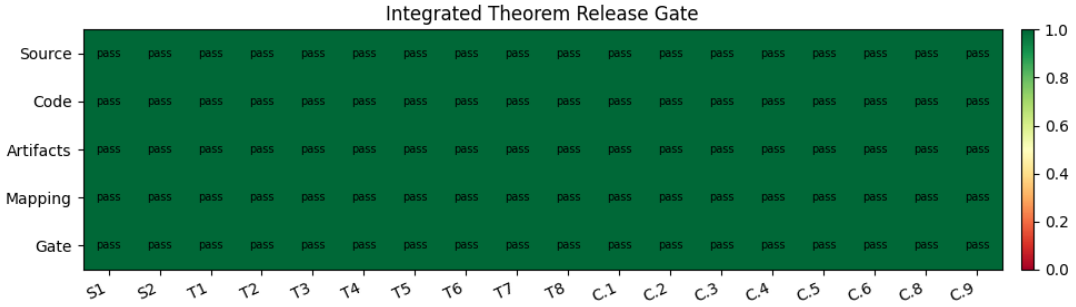

Figure 5.1: The integrated theorem gate keeps theorem language aligned to locked code and artifact surfaces rather than to informal prose summaries.

The dissertation repeatedly returns to one discipline rule: theorem statements must point to explicit assumptions, typed runtime objects, and tracked empirical diagnostics. That is why the proof surface remains durable even when empirical maturity differs across domains.

5.5 Proved, validated, and roadmap claims

The theory chapter is also the place where the dissertation's claim boundary becomes reader-visible. Proved claims live here and in the proof appendix. Validated claims are only promoted when replay, calibration, runtime, and governance artifacts exist. Roadmap claims remain in bounded future-work language. This separation is the mechanism that lets ORIUS argue for a universal safety layer without pretending that every supported row is already equal in empirical closure.

Chapter 6

Benchmark and Reproducibility Protocol

6.1 Dual-trajectory evaluation as the core benchmark principle

ORIUS-Bench is built around dual-trajectory replay. Every evaluation episode distinguishes between what the runtime *observes* and what the plant or replay harness treats as the *true* safety surface. This is the only reliable way to measure OASG, because observed-state legality alone cannot reveal whether the action was physically safe.

6.2 Replay, faults, and deterministic schedules

The benchmark uses deterministic fault schedules so that replay, comparison, and certificate traces remain reproducible. Fault families include dropout, stale holds, delay, reordering, spikes, drift, and cross-domain analogs. The important scientific point is not merely that faults exist; it is that they can be replayed against the same true-vs-observed contract across domains.

6.3 ORIUS-Bench metrics

The dissertation evaluates ORIUS through a metric family that matches the runtime object: true-state violation rate (TSVR), observation-action safety gap, certificate validity, graceful degradation quality, intervention burden, audit completeness, and runtime latency.

These metrics prevent the dissertation from drifting into a pure forecasting or pure control performance narrative.

Table 6.1: Canonical universal replay output. Battery is the locked reference row; industrial and healthcare are proof-validated; AV is bounded proof-validated under the TTC contract; navigation is shadow-synthetic; aerospace is experimental.

Domain	Faults	Baseline TSVR	ORIUS TSVR	Reduction (%)
battery	multi	0.75	0.875	-16.7
navigation	multi	0.6667	0.25	62.5
industrial	multi	1.0	0.0	100.0
healthcare	multi	1.0	0.5833	41.7
aerospace	multi	1.0	0.0	100.0
vehicle	multi	0.0833	0.0	100.0

6.4 Artifact provenance and claim locking

The benchmark protocol is inseparable from the claim-lock pipeline. The canonical thesis surface is `paper/paper.tex`, but its headline tables and figures are backed by manifests, publication matrices, and governed artifact paths. This means the dissertation is not "about" results in the abstract; it is a storyboard of verifiable evidence.

6.5 Strict universal mode

For the dissertation, benchmark discipline also means refusing battery-shaped shortcuts in universal language. Domain-neutral replay predicates, adapter hooks, and parity gating are the canonical surfaces. Legacy compatibility fields can remain for historical or bounded support use, but they are not allowed to define the universal benchmark story.

Part III

Witness Domain Evidence

Chapter 7

Witness Domain Implementation

7.1 Why battery is the witness domain

Battery remains the deepest witness row because it carries the strongest theorem-to-code-to-artifact lineage in the repository. That makes it the right place to show the full ORIUS stack end to end without pretending that the dissertation is only about energy systems. The witness-domain role is therefore methodological: battery is the row where the runtime contract, theorem ladder, replay protocol, and claim-lock pipeline are all simultaneously visible.

7.2 Data and dataset cards

The battery surface is built from tracked regional energy and telemetry artifacts rather than from a single benchmark file committed to git. The dissertation keeps the data story artifact-first and manifest-driven so that large external datasets can still participate in a reproducible thesis pipeline.

Table 7.1: Dataset-card inventory for the conference release family.

dataset_key	dataset_label	country	rows	date_start	date_end	targets	min_coverage_pct	feature_count	columns	source	weather_source	carbon_source
DE	Germany (OPSD)	DE	17377	2018-10-07	2020-09-30	load_mw,solar_mw,wind_mw	100.000000	94	98	OPSD + SMARD	Open-Meteo archive API	SMARD generation mix
US_ERCOT	USA (EIA-930 ERCOT)	US	13453	2024-07-01	2026-01-20	load_mw,solar_mw,wind_mw	100.000000	62	62	EIA-930 ERCOT	Open-Meteo archive API	proxy_generation_mix
US_MISO	USA (EIA-930 MISO)	US	13638	2024-07-01	2026-01-20	load_mw,solar_mw,wind_mw	100.000000	114	118	EIA-930 MISO	Open-Meteo archive API	proxy_generation_mix
US_PJM	USA (EIA-930 PJM)	US	13639	2024-07-01	2026-01-20	load_mw,solar_mw,wind_mw	100.000000	62	62	EIA-930 PJM	Open-Meteo archive API	proxy_generation_mix

7.3 Forecasting, calibration, and reliability surfaces

The witness row deliberately separates the engineered forecasting track from the degraded-observation novelty track. The current engineered track remains a strong systems baseline,

while the learned-reliability and raw-sequence surfaces are framed as bounded deep-learning novelty inside ORIUS rather than as a replacement thesis identity.

Table 7.2: Promoted six-model forecasting comparison. Deep learning baselines evaluate point-forecast performance, confirming GBM's operational dominance in this tabular regime. Conformal uncertainty artifacts (PICP, Width) are materialized exclusively for the production-candidate model (GBM).. *— = n/a; PICP@90 and Width only for GBM.*

Region	Target	Model	RMSE	MAE	sMAPE (%)	R^2	PICP@90	Width (MW)
DE	Load	**GBM**	**305.12**	**187.87**	**0.39**	**0.9988**	88.2	1070.2
DE	Load	LSTM	—	—	—	—	—	—
DE	Load	TCN	—	—	—	—	—	—
DE	Load	N-BEATS	3646.28	2897.93	5.84	0.8347	—	—
DE	Load	TFT	8996.05	7892.52	15.52	-0.0060	—	—
DE	Load	PatchTST	4087.86	3142.77	6.22	0.7923	—	—
DE	Wind	**GBM**	**219.11**	**129.36**	**2.41**	**0.9991**	93.7	833.9
DE	Wind	LSTM	—	—	—	—	—	—
DE	Wind	TCN	—	—	—	—	—	—
DE	Wind	N-BEATS	7571.15	6232.12	67.76	-0.0504	—	—
DE	Wind	TFT	7312.89	5540.71	63.80	0.0200	—	—
DE	Wind	PatchTST	7383.74	5974.24	66.52	0.0009	—	—
DE	Solar	**GBM**	**240.35**	**117.32**	**69.24**	**0.9993**	78.8	1037.7
DE	Solar	LSTM	—	—	—	—	—	—
DE	Solar	TCN	—	—	—	—	—	—
DE	Solar	N-BEATS	3441.45	2192.56	104.80	0.8553	—	—
DE	Solar	TFT	9067.28	8041.60	128.70	-0.0042	—	—
DE	Solar	PatchTST	3136.94	2030.22	102.24	0.8798	—	—

Table 7.3: Battery DeepOQE diagnostics on the held-out degraded-telemetry episodes.

Metric	Heuristic	DeepOQE
Reliability MAE	0.1176	0.1622
Fault AUC	—	0.5345
Low-reliability fault lift	—	0.000

7.4 Control baselines and adapter implementation

Battery is also the cleanest row for showing how the adapter binds state semantics, uncertainty construction, action repair, fallback, and certificate emission to a real plant envelope. The dissertation uses that row to explain the implementation discipline: nominal forecasting and control are replaceable, but the ORIUS adapter boundary and replay contract remain fixed.

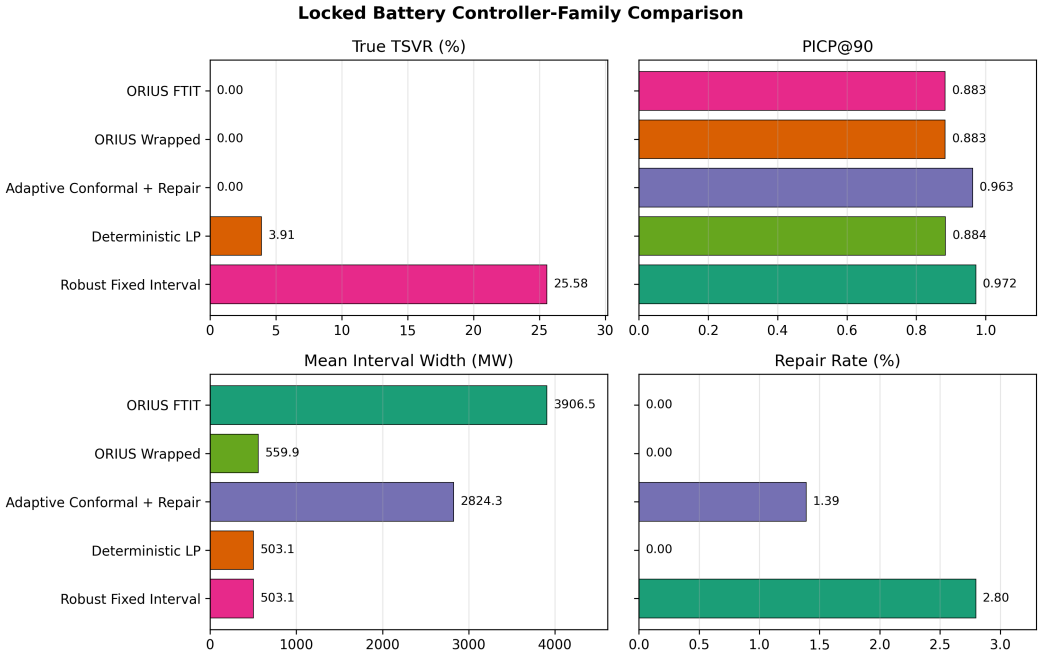

Figure 7.1: Witness-domain reliability baselines. The engineered forecasting surface and the learned reliability surface are intentionally reported together so the dissertation does not confuse forecasting novelty with degraded-observation novelty.

7.5 Locked artifact protocol

The witness row is defended through locked artifacts rather than through narrative memory. Training surfaces, replay outputs, theorem traces, calibration diagnostics, and runtime tables all point back to generated publication assets or manifests. This is the point at which the chapter-first dissertation model and the repository's claim-lock machinery meet: the prose is curated, but the evidence remains machine-governed.

Chapter 8

Witness Results and Failure Analysis

8.1 The governing safety question

The battery witness row answers one governing question for the dissertation: when telemetry degrades, can a runtime layer reduce true-state safety loss without pretending that nominal controller legality is enough? This chapter treats the battery row as the deepest defended empirical surface because it closes the full ORIUS loop from degraded observation, to repaired action, to governed release.

8.2 Witness experiment design

The witness experiments are not meant to show that ORIUS dominates every plant-specific controller under every cost metric. They are designed to test a narrower and more important claim: whether a certificate-aware runtime can keep hidden true-state violations from passing through a nominally coherent observation surface. The baselines therefore include controllers that are cheaper, controllers that are more aggressive, and controllers that already include some robustness logic, because the scientific question is not merely who optimizes best in nominal conditions. The question is what survives once the observation channel itself begins to fail.

8.3 Headline result

The primary evidence surface is the main safety-and-cost table below.

The table shows the core ORIUS result in the form that matters most for this dissertation. Several plant-specific robust baselines already avoid true-state violations on this witness

Table 8.1: Main safety and cost outcomes under telemetry faults.

controller	true_soc_violation_rate_mean	true_soc_violation_rate_ci_low	true_soc_violation_rate_ci_high	true_soc_violation_severity_p95_mean	true_soc_violation_severity_p95_ci_low	true_soc_violation_severity_p95_ci_high	intervention_rate_mean	intervention_rate_ci_low	intervention_rate_ci_high	cost_delta_pct_mean
aci_conformal	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.020833	0.020833	0.020833	0.023244
cvar_interval	0.191667	0.119427	0.274305	333.582626	0.265437	848.045446	0.000000	0.000000	0.000000	-0.009092
dcls_fixi	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.088194	0.070139	0.105556	0.147639
dcls_wrapped	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.054576
deterministic_lp	0.129861	0.105555	0.170833	547.992608	198.317264	1056.208862	0.000000	0.000000	0.000000	0.000000
robust_fixed_interval	0.191667	0.119427	0.274305	333.582626	0.265437	848.045446	0.000000	0.000000	0.000000	-0.009092
scenario_mpc	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.006481
scenario_robust	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.006481

task, so the contribution is not that ORIUS is the only safe controller in the table. The contribution is that ORIUS closes the hidden-violation mode while keeping intervention, repair, and certificate semantics explicit at runtime. In contrast, the deterministic and interval baselines still exhibit material true-state violation rates once telemetry faults are introduced, even though they remain internally well-posed optimization procedures.

That distinction is the witness-domain result. ORIUS is not claiming to replace every robust controller with one universal optimizer. It is showing that a runtime layer can sit between nominal control and physical release, detect when observation quality has changed the meaning of legality, and then either repair the action or deny release under explicit certificate semantics.

8.4 Why the result happens

The battery witness supports a mechanistic interpretation rather than a leaderboard reading. ORIUS changes the release decision in three linked ways. Reliability loss is detected before action release, uncertainty is widened before legality is checked, and unsafe actions are either repaired or replaced before they reach the plant. The main table is therefore best read as evidence that the release boundary changed, not simply as evidence that one model forecasted more accurately than another.

8.5 Ablation logic

The ablation table shows where the runtime stack earns its gains and where those gains still depend on the details of the degraded-observation regime.

Some secondary scenarios clear the strict gate and others do not. That pattern matters scientifically because it shows where the kernel is buying safety: not through a single magic margin, but through a coupling between reliability estimation, uncertainty widening, and local repair geometry. The ablations therefore support a sharper claim than simple average improvement: ORIUS behaves like a genuine degraded-observation runtime layer whose effect depends on the structure of the fault family.

Table 8.2: Ablation study with strict gate fields and statistical tests.

Scope	Fault	n	Viol. base	Viol. DC3S	Viol. red.	p_v	Sev. base	Sev. DC3S	Sev. red.	p_s	Pass
primary	all	160	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	delay	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	dropout	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	out_of_order	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	spikes	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary_scenario	aggregate	10	0.35	0.30	14.3	0.002	0.35	0.30	14.3	0.002	yes
secondary_scenario	aggregate	10	0.25	0.30	-20.0	0.002	0.25	0.30	-20.0	0.002	no
secondary_scenario	aggregate	10	0.35	0.30	14.3	0.002	0.35	0.30	14.3	0.002	yes
secondary_scenario	aggregate	10	0.25	0.30	-20.0	0.002	0.25	0.30	-20.0	0.002	no

8.6 Coverage under reliability slices

ORIUS depends on reliability-conditioned uncertainty rather than on a single fixed interval. The grouped coverage table below is included to show why that distinction matters.

Table 8.3: Regime-group CQR coverage and interval width summary.

Target	Regime	PICP@90 (%)	Width (MW)	n
Load	LOW	85.2	1244.8	840
Load	MED	88.3	937.4	840
Load	HIGH	91.1	1028.2	840
Wind	LOW	97.6	719.7	840
Wind	MED	93.5	812.0	840
Wind	HIGH	90.0	969.9	840
Solar	LOW	80.2	891.6	840
Solar	MED	82.2	1016.6	845
Solar	HIGH	73.8	1206.2	835

Coverage quality moves materially across regimes, especially in the harder operating slices. That is exactly the pattern ORIUS is designed to take seriously. If interval behavior were uniform, a fixed robust margin would be closer to sufficient. Because interval behavior changes with regime and reliability, the runtime needs to tighten admissible action as trust in the observation surface falls.

8.7 Bounded novelty inside the witness row

The witness row also contains a bounded learned-reliability novelty surface. It matters here only insofar as it improves the degraded-observation safety argument rather than turning the dissertation into a forecasting paper.

The figure supports a narrower point than the main safety table. It shows that the learned reliability path can improve the runtime's discrimination of when to intervene and when to

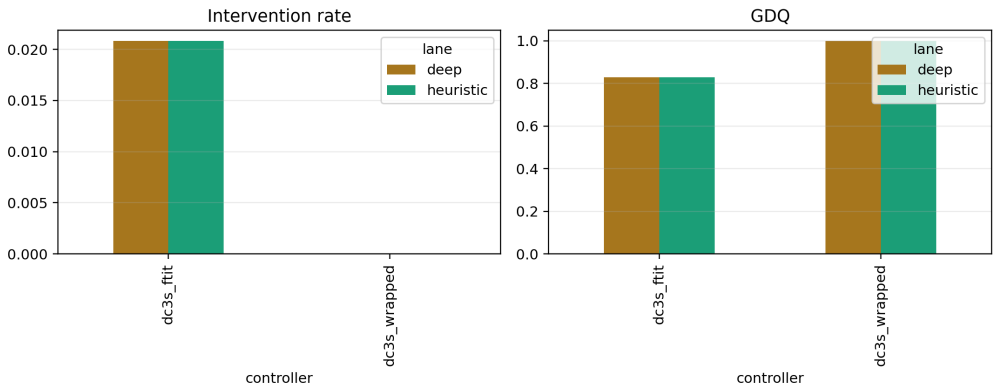

Figure 8.1: Battery learned-reliability safety surface. The deep model is judged by true-state safety, intervention burden, and certificate-valid release behavior rather than by forecast error alone.

allow nominal action to pass. That is a bounded novelty claim about the observation side of ORIUS, not a replacement for the main witness claim about repaired release under degraded telemetry.

8.8 Without ORIUS

Without ORIUS, the telemetry fault is treated as a nuisance external to the release decision. The deterministic and interval baselines in the headline table expose exactly what goes wrong under that assumption: the controller remains computationally coherent while true-state violation reappears. Without calibrated widening, the controller checks legality against the wrong state set. Without repair, the unsafe action is still released. Without certificate discipline, the runtime cannot later explain why the action was allowed or when fallback should have taken over.

8.9 What still remains bounded

The battery chapter therefore supports three distinct claims. First, it supports a strong witness claim that ORIUS can materially improve degraded-observation safety on a defended row. Second, it supports a bounded robustness claim that the kernel remains meaningful across multiple telemetry fault families rather than on one curated failure. Third, it supports a runtime-governance claim that intervention and certificate behavior are part of the result, not merely surrounding instrumentation. Deeper learned-reliability artifacts remain in the governed package, but the main witness conclusion is already visible here: once legality on

the observed state is no longer enough, ORIUS changes the release boundary in a measurable and auditable way.

Part IV

Universalization and Extensions

Chapter 9

Universal ORIUS Across Domains

9.1 What universality means in this dissertation

ORIUS is universal in a specific sense. It does not assume that batteries, vehicles, industrial plants, healthcare systems, navigation stacks, and aerospace controllers share a plant model, control law, or optimization target. It assumes only that each domain must answer the same release-boundary question once observation becomes unreliable: what action may still be released safely, under what margin, with what fallback, and under what certificate? The shared object is therefore not the nominal controller. It is the runtime safety logic that mediates degraded observation before actuation.

9.2 The shared domain template

Every domain row in this dissertation is described through the same template:

1. the local degraded-observation hazard;
2. the ORIUS instantiation wrapped around nominal intelligence;
3. the defended safety object;
4. the current evidence status; and
5. the universality contribution that row makes to the thesis.

That template is what makes cross-domain comparison scientific rather than rhetorical. The rows are not interchangeable, but they are comparable because ORIUS forces them to answer the same runtime question through one implementation grammar: a `DomainAdapter`

binding, one Detect–Calibrate–Constrain–Shield–Certify step, one benchmark schema, and one certificate-aware release semantics.

9.3 Cross-domain status at a glance

Table 9.1: Equal-domain parity gate for the universal-first ORIUS book.

Domain	Adapter	Raw/Data	Train	Replay	Soundness	Fallback	CertOS	P5	Tier / Exact blocker
Battery Energy Storage	pass	locked_reference	pass	pass	reference_witness	paper6_runtime	evaluated	evaluated	reference; battery_reference_witness
Autonomous Vehicles	pass	verified	pass	pass	pass	bounded_runtime_pass	evaluated	gated	proof_validated; promoted
Industrial Process Control	pass	verified	pass	pass	pass	bounded_runtime_pass	evaluated	evaluated	proof_validated; industrial_train_validation_chain_complete
Medical and Healthcare Monitoring	pass	verified	pass	pass	pass	bounded_runtime_pass	evaluated	gated	proof_validated; healthcare_train_validation_chain_complete
Navigation and Guidance	pass	blocked	blocked_real_data_gap	pass	blocked_real_data_gap	bounded_runtime_pass	gated	gated	shadow_synthetic; navigation_real_data_gap
Aerospace Control	pass	placeholder_synthetic	placeholder_surface	pass	experimental_placeholder	bounded_runtime_pass	gated	gated	experimental; aerospace_experimental_placeholder

The parity matrix summarizes the current empirical boundary of the dissertation. Battery is the witness row. Industrial process control and healthcare monitoring are defended bounded rows. The autonomous-vehicle row is a defended bounded runtime row under the current artifact surface, while canonical raw-data closure remains incomplete. Navigation remains shadow-synthetic, and aerospace remains experimental. The matrix should therefore be read as the evidence boundary for a universal runtime architecture, not as proof that every row has already reached equal maturity.

9.4 Domain synthesis

Battery energy storage. The battery hazard is hidden constraint violation under degraded feeder, load, or state telemetry. ORIUS instantiates the runtime with tight state margins, explicit repair, and auditable release over a heavily instrumented control surface. Its evidence status is the deepest in the dissertation, which is why battery serves as the witness row. Its universality contribution is to show that the full theorem-to-code-to-artifact chain can be closed on a real physical control problem.

Autonomous vehicles. The autonomous-vehicle hazard is unsafe longitudinal release under stale or degraded scene understanding. ORIUS instantiates the runtime through bounded time-to-collision and predictive-entry-barrier logic, with explicit brake-oriented fallback and certificate-aware release checks. Its evidence status is a defended bounded runtime row under the current artifact surface, while canonical raw-data closure remains incomplete. Its universality contribution is to show that the same release semantics remain meaningful in a fast-horizon, continuous-action mobility setting.

Industrial process control. The industrial hazard is constraint excursion under corrupted or delayed plant telemetry. ORIUS instantiates the runtime through bounded power or process envelopes, repair against shared operating constraints, and governed fallback. Its evidence status is a defended bounded row. Its universality contribution is to show that ORIUS can operate in tightly regulated process loops where plant constraints are explicit and intervention must remain operationally interpretable.

Healthcare monitoring. The healthcare hazard is clinically unsafe release under partially observed physiological state, especially when alerts and interventions are triggered from noisy or degraded streams. ORIUS instantiates the runtime through physiological margin checks, alert fallback, and certificate-aware release semantics. Its evidence status is a defended bounded row. Its universality contribution is to show that the ORIUS grammar survives in a setting where the protected object is patient-facing monitoring behavior rather than actuation on an energy or motion platform.

Navigation and guidance. The navigation hazard is trajectory release under degraded localization or scene reference. ORIUS instantiates the runtime through hold-position or corridor-preserving fallback logic. Its current evidence status remains shadow-synthetic because the defended real-data train–replay chain is not yet closed. Its universality contribution is still valuable: it shows how the ORIUS release boundary maps onto spatial feasibility problems even before the row is fully defended.

Aerospace control. The aerospace hazard is unsafe envelope or guidance release under degraded flight observation. ORIUS instantiates the runtime through envelope-aware repair, bounded fallback, and certificate semantics tied to flight safety objects. Its current evidence status remains experimental because a defended multi-flight runtime validation surface is still missing. Its universality contribution is to show that ORIUS extends to high-consequence control domains without pretending that structural portability already implies empirical closure.

The figure should be read as an interpretation aid rather than as a replacement for the parity table. It makes visible that the same runtime object appears in every domain, even though the domains differ sharply in dynamics, sensing, and safety geometry.

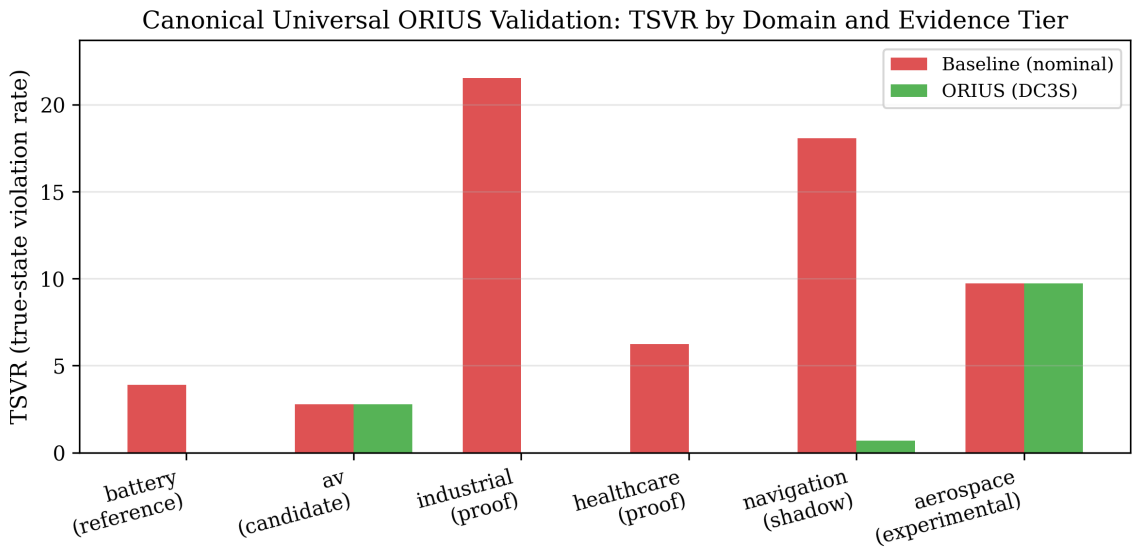

Figure 9.1: Cross-domain ORIUS validation surface. One runtime grammar carries across six domains, while each row remains limited by its own defended evidence surface.

9.5 Cross-domain interpretation

Two conclusions follow from the domain synthesis. First, what transfers is not a controller family but a release-boundary grammar: reliability loss, uncertainty inflation, action repair, fallback semantics, and certificate-aware release. Second, what remains local is the geometry of the safety object itself: a battery margin is not a vehicle headway constraint, and a vehicle headway constraint is not a physiological alert threshold or a flight envelope. ORIUS matters scientifically because it preserves one runtime meaning while allowing those local geometries to remain local.

The most important implication is that universality here is structural rather than homogenizing. ORIUS does not make domains identical. It makes them legible under one safety question: when reality and observation diverge, what action may still be released, and with what certificate? That is why a battery witness row, a bounded vehicle row, industrial and healthcare rows, and still-gated navigation and aerospace rows can all belong in one dissertation without collapsing into a single benchmark story.

9.6 Blocked rows and the present thesis boundary

The blocked rows are simple to state and should remain visible. Navigation is not yet a defended peer because the real-data closure row is still missing. Aerospace is not yet a defended peer because the dissertation does not yet contain a defended multi-flight runtime

surface with material post-repair gain. That boundary does not weaken the universal thesis. It clarifies it: ORIUS already establishes a reusable runtime safety grammar across domains, while equal-domain closure remains an empirical task rather than a rhetorical shortcut.

Chapter 10

Temporal Certificates and Graceful Degradation

10.1 Certificate horizon as a runtime object

ORIUS certificates are temporal objects, not merely one-step labels. A certificate has a validity horizon, an expiration condition, and a bounded interpretation under worsening observation. This matters because degraded telemetry can remain acceptable for one step and become unacceptable only a short time later. The temporal question is therefore not "safe or unsafe forever," but "how long does the runtime have before the current certificate no longer justifies release?"

10.2 Expiration, half-life, and safe-hold

The dissertation treats horizon and half-life conservatively. The current evidence is strongest for bounded expiration and safe-hold semantics under blackout-style degradation. That is the right level of claim: certificate time is measurable and governable, but it is not yet promoted into a universal law of every plant and every sensor regime.

10.3 Graceful degradation policies

When no full-quality control action remains justified, ORIUS benchmarks degradation policies explicitly rather than treating them as informal emergency behavior. This is the runtime-assurance connection in practice: the system needs a governed way to do less, safer, and more explainably when observation quality collapses.

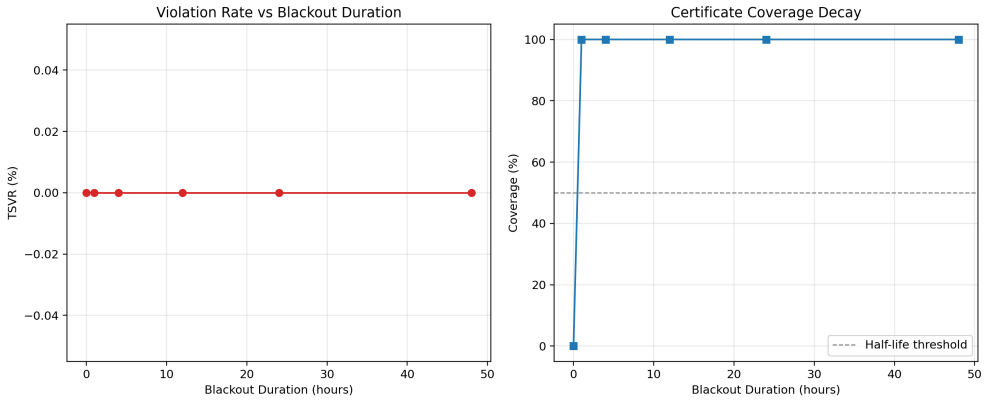

Figure 10.1: Blackout half-life surface used to motivate certificate expiration, bounded safe-hold, and temporal intervention discipline.

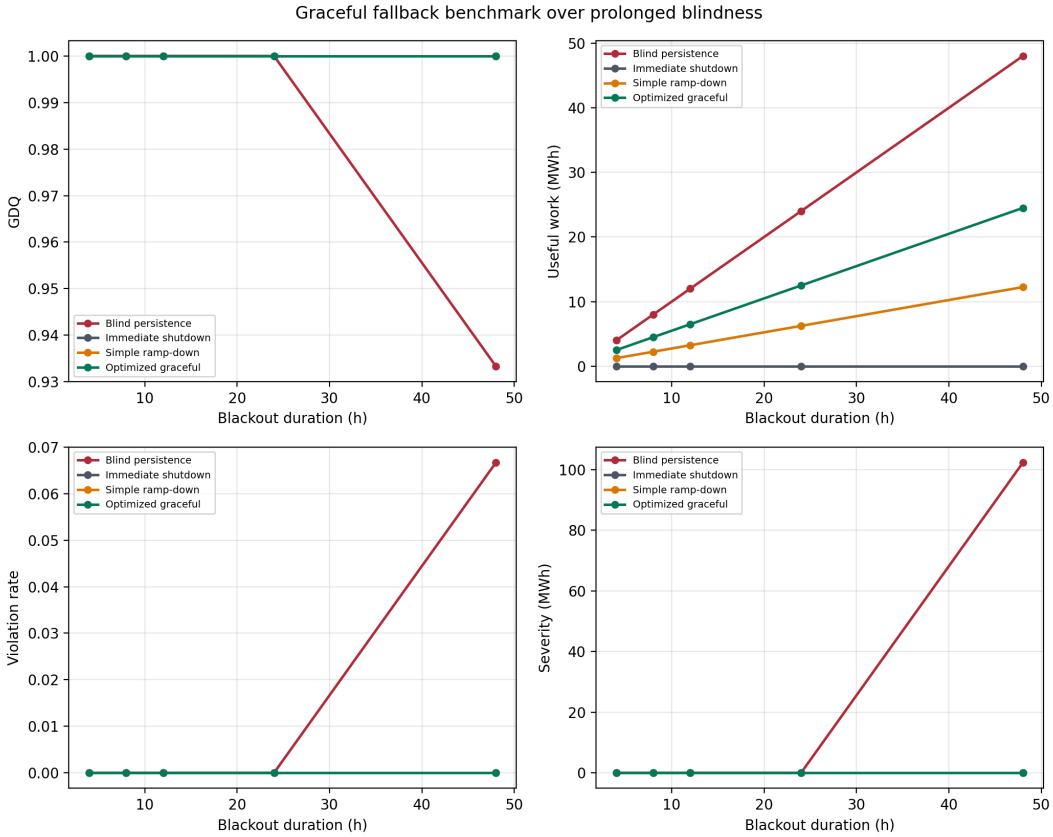

Figure 10.2: Four-policy graceful degradation comparison. The dissertation uses this family to treat fallback quality as a benchmarked layer instead of an unexamined emergency default.

10.4 Bounded claim

The temporal chapter supports a bounded but important claim: certificate time and graceful degradation can be benchmarked and governed as part of the same runtime object as repair and fallback. It does not claim a universal optimal degradation policy, and it does not erase the domain-specific obligations that remain tied to the parity matrix.

Chapter 11

Compositional Safety

11.1 Why local safety does not automatically compose

One of the easiest mistakes in safety argumentation is to assume that individually safe controllers remain jointly safe when they share resources or constraints. ORIUS treats composition as a bounded extension precisely because this implication fails in general. Local certificates can remain valid while shared budgets, timing windows, or coupled plant dynamics still create system-level failure.

11.2 Shared constraints and negotiated safety

The dissertation therefore frames composition around shared constraints: fleet power budgets, corridor occupancy, shared actuator authority, or other common resources. The point is not to claim a universal composition theorem today. It is to show that ORIUS can lift the certificate object into a negotiated or multi-agent setting without pretending that local repair semantics are automatically sufficient.

Table 11.1: Multi-Agent Fleet Coordination: joint safety violations and useful work under shared transformer capacity constraint (limit = 80 MW, two batteries at 60 MW each = 120 MW proposed). Independent protocol violates; ORIUS-coordinated eliminates violations.

Protocol	Joint Violations	Local Violations	Useful Work (MWh)	Margin Quality
Independent (non-compositional)	24	0	1920	0
Centralized (ORIUS-coordinated)	**0**	0	1920	1
Distributed Negotiation	**0**	0	1920	1

11.3 Bounded extension rather than universal closure

Composition remains a defended bounded layer in the dissertation. It matters scientifically because it shows that the runtime grammar does not end at single-agent control. At the same time, it remains outside the strongest universal claim because cross-domain shared constraints still require more defended rows before equal-domain composition language would be justified.

Chapter 12

CertOS Runtime Assurance

12.1 CertOS as release semantics

CertOS is the part of ORIUS that turns repaired action into a governed runtime object. Without lifecycle state, fallback rules, and an audit chain, certificates would be inert metadata. With CertOS, certificate validity becomes part of the release decision itself: which action was allowed, under which reliability state, for which horizon, with which fallback if the certificate expires or becomes invalid?

This is the point at which the ORIUS kernel acquires operational semantics. `DomainAdapter` defines the local safety predicate and fallback object that CertOS must protect, and the universal benchmark schema defines the replay traces from which certificate validity, latency, and lifecycle behavior are evaluated. In that sense, CertOS is the computational bridge between repaired action and a safety argument that can later be audited.

12.2 Lifecycle invariants

The lifecycle logic is narrow and strong. A protected action is never released without a valid certificate or an explicit fallback. Certificate state must be renewable, expirable, revocable, and auditable. Runtime traces must preserve enough information to reconstruct why the release was allowed and when the governing assumptions ceased to hold. These are not administrative conveniences. They are the invariants that keep repaired action tied to a reviewable safety argument.

The figure highlights the right scientific point: governance is not separate from runtime safety once release depends on repaired action under degraded observation. What varies across rows is the maturity of the evaluated lifecycle surface, not the meaning of the lifecycle

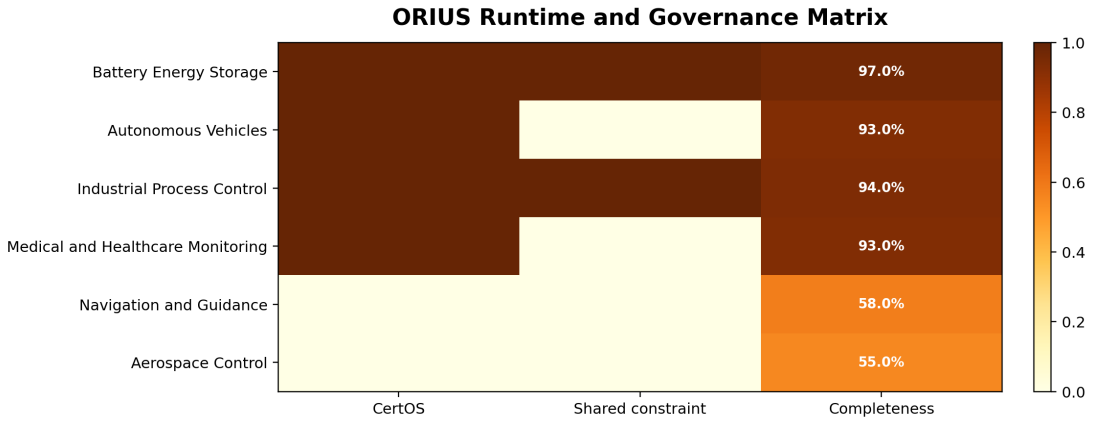

Figure 12.1: CertOS lifecycle and runtime-governance surface across ORIUS domains. The figure shows how certificate issue, validation, expiry, fallback, and audit continuity are exposed as part of runtime safety rather than as post hoc reporting.

itself.

12.3 Architecture-level universality, evidence-tiered maturity

CertOS is universal at the level of release semantics. Every defended ORIUS row needs a certificate object, fallback policy, and audit trace. What is not yet universal is equal deployment maturity across every domain. Battery, industrial process control, and healthcare provide the strongest defended lifecycle surfaces. The autonomous-vehicle row shows a defended bounded runtime surface under current artifacts. Navigation and aerospace remain gated for the same empirical reasons already stated elsewhere in the dissertation. CertOS therefore supplies one common governance object without pretending that all deployment surfaces are equally mature today.

12.4 Exact governance claim

The CertOS claim is exact and intentionally narrower than a flat deployment slogan: ORIUS provides a policy-driven runtime governance layer in which repaired action, fallback, and certificate validity are coupled, auditable, and recoverable after the fact. That claim is scientifically important because a runtime safety layer without release semantics cannot say what action was justified once degraded observation has already entered the control loop.

Part V

Submission Boundary

Chapter 13

Governance, Reproducibility, Limitations, and Conclusion

13.1 Limitations and blocked rows

The dissertation's strongest limitation is empirical, not conceptual. The runtime grammar is broader than the deepest currently defended evidence. Battery remains the witness row with the most complete theorem-to-code-to-artifact chain. Industrial process control and healthcare monitoring support defended bounded interpretations. The autonomous-vehicle row is a defended bounded runtime row under the current artifact surface, while canonical raw-data closure remains incomplete. Navigation remains blocked by the absence of a defended real-data train–replay chain. Aerospace remains blocked by the absence of a defended multi-flight runtime validation surface with material post-repair gain.

Those limits matter because ORIUS is not presented here as a universal optimal controller or as a finished equal-domain deployment program. It is presented as a runtime safety layer whose strongest current evidence is deep in one witness domain and bounded across several others. The open rows should therefore be read as the remaining empirical work required to extend the thesis, not as defects that invalidate the defended core.

13.2 Reproducibility and evidence boundary

The manuscript keeps every headline claim tied to tracked figures, tables, manifests, or claim matrices so that the defended surface remains auditable after the fact. The purpose of that discipline is simple: a universal safety thesis should be inspectable at the level of artifacts, not only persuasive at the level of prose. In that exact sense, the dissertation is universal in

architecture, tiered in evidence.

The evidence boundary is therefore factual rather than rhetorical. Claims backed by theorem statements remain theorem claims. Claims backed by replay, runtime traces, and governed artifacts remain validated claims. Claims that depend on future real-data closure for navigation or multi-flight aerospace closure remain open research steps and are not promoted into defended headline language.

13.3 Conclusion

ORIUS establishes one durable idea for physical AI: degraded observation should be treated as a first-class runtime safety problem rather than as a residual modeling nuisance. Whenever observed-state legality and true-state legality can diverge, safe release requires more than nominal intelligence. It requires explicit reliability estimation, conservative state widening, repaired action or bounded fallback, and certificate-aware release.

The mechanism that makes that claim concrete is explicit in the dissertation. A `DomainAdapter` exposes the domain safety predicate and fallback object. The universal benchmark schema exposes one replay contract for truth, observation, intervention, fallback, certificate validity, and latency. CertOS exposes the lifecycle semantics that determine when repaired action may still be released. Together, those three surfaces are the implementation backbone of ORIUS and the reason this thesis is more than a conceptual safety manifesto.

ORIUS therefore does not ask every domain to share a controller, a plant model, or an optimization objective. It asks every domain to make the release boundary explicit. By doing so, it turns true-state violation, intervention burden, fallback usage, certificate validity, and runtime latency into scientific objects that can be compared across domains without pretending the domains themselves are identical.

The thesis therefore closes on a claim that is both ambitious and disciplined. ORIUS already supports a reusable runtime safety layer for physical AI under degraded observation. It does not yet support the stronger claim that every supported domain is equally mature. That boundary should be treated as a scientific virtue rather than a rhetorical compromise: structural universality is defended now, while equal-domain closure remains an empirical program that must be earned.

If physical AI is to move beyond domain-by-domain reactive patching, it needs a common runtime discipline for deciding what remains safe when the observation channel begins to fail. ORIUS provides that discipline in a form that is mathematical enough to be analyzed, empirical enough to be tested, and operational enough to govern release. That is the thesis claim this dissertation defends.

Appendix a

Assumption Register

This appendix records the assumptions that delimit the dissertation's strongest claims. The purpose is not to hide caveats in the back matter. It is to make the theorem and evidence boundary explicit enough that every promoted claim can be traced to the assumptions it needs.

Table a.1: ORIUS dissertation assumption register.

ID	Name	Meaning	Where it matters
A1	Bounded observation error surface	Reliability-conditioned uncertainty inflation produces a finite observation-consistent state set.	Chapters 3, 5, and the benchmark appendix.
A2	Feasible safe repair or fallback	When the admissible set collapses, the runtime can still choose a certified fallback action or safe-hold policy.	Chapters 4, 10, and 12.
A3	Causal certificate issuance	Certificates depend only on information available at runtime up to the current step.	Chapters 4, 12, and the safety-case appendix.

ID	Name	Meaning	Where it matters
A4	Dual-trajectory replay fidelity	The benchmark exposes both observed and true safety surfaces strongly enough to measure OASG.	Chapters 6, 8, and 9.
A5	Adapter faithfulness	Each domain adapter encodes the relevant safety predicate, fallback semantics, and certificate payload honestly for its plant.	Chapters 4, 9, and 12.
A6	Evidence-tier discipline	Architectural universality and empirical parity are not allowed to collapse into one claim.	Chapters 1, 9, and 13.

The dissertation's formal claims should be read together with this table. When the evidence surface is weaker than the architectural surface, the text is intentionally written as a bounded semantic or governance statement rather than as an equal-domain validation claim.

Appendix b

Proofs

This appendix collects the proof-facing surface of the dissertation at the level required for manuscript traceability. The full repository still contains a richer battery-era proof archive, but the canonical dissertation proof surface is the theorem ladder summarized here, tied directly to the new chapter flow.

b.1 Theorem ladder restatement

1. **OASG existence.** Under degraded observation, an action can be safe on the observed state and unsafe on the true state.

2. **Safety preservation under repair.** If the repaired action lies inside the admissible action set generated from the observation-consistent state set, then true-state safety is preserved under the stated assumptions.

3. **Degradation-sensitive risk envelope.** As reliability decreases, the uncertainty set expands and the admissible action budget contracts.

4. **No-free-safety principle.** A quality-ignorant controller cannot guarantee zero true-state safety violation under admissible degraded-observation regimes.

b.2 Proof skeletons

OASG existence Let $\hat{x}_t$ be the observed state and $x_t^\star$ the true state. If telemetry degradation makes the observation error nonzero while the nominal controller evaluates legality on $\hat{x}_t$ alone, then there exist states for which $u_t \in \mathcal{C}(\hat{x}_t)$ but $u_t \notin \mathcal{C}(x_t^\star)$. The proof follows by

constructing a degradation regime that moves the true state across a safety boundary while preserving apparent legality on the observed state.

Safety preservation under repair ORIUS avoids that event by replacing the point observation with an observation-consistent set $\mathcal{X}_t$. Repair then projects the candidate action into an admissible set defined over $\mathcal{X}_t$. If the repair operator is feasible or an explicit fallback exists, release of the repaired action preserves the defended safety predicate under the assumed uncertainty and dynamics bounds.

Risk-envelope monotonicity The risk-envelope argument follows from monotone inflation: worse reliability cannot justify a smaller state set or a larger admissible action set. This creates a law-like relationship between degraded telemetry and intervention pressure, which is why intervention is a scientific metric in ORIUS rather than a purely operational nuisance.

No-free-safety If a controller ignores observation quality, then an admissible degradation process can always be constructed that makes its fixed action semantics unsafe on the true state. The runtime layer is therefore not optional decoration; it is the mechanism that closes the gap between nominal control and physical legality.

b.3 Proof-to-artifact linkage

Theorems in the dissertation are promoted only when they have both a formal statement and a matching artifact surface:

- theorem dependency table in Chapter 5
- integrated theorem gate figure in Chapter 5
- witness-domain replay and calibration evidence in Chapters 7 and 8
- parity and claim-boundary surfaces in Chapters 9 and 13

This is the dissertation's proof discipline: theorem statements, replay evidence, and claim promotion remain coupled without being conflated.

Appendix c

Benchmark Specifications

This appendix records the benchmark-facing specifications that make the dissertation replayable and auditable. ORIUS-Bench is universal at the schema level: every episode tracks candidate action, repaired action, true-state violation, observed-state satisfaction, certificate state, fallback/intervention markers, and runtime cost. The current deepest empirical instantiation remains the battery witness row, so the detailed fault-family table below is still battery-grounded. That grounding is deliberate: it shows the strongest defended fault surface while the benchmark contract itself remains domain-neutral.

c.1 Canonical fault families

Table c.1 defines each fault family, its configurable parameters, the tested range, and the battery-domain interpretation.

Table c.1: Witness-domain fault families used to instantiate the universal ORIUS-Bench telemetry-degradation contract.

Fault	**Parameters**	**Range tested**	**Battery interpretation**
`dropout`	drop_rate d	$d \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$	Packet loss on the BMS→EMS telemetry link. Dropped readings are replaced by the last received value (zero-order hold).

(continued)

Fault	Parameters	Range tested	Battery interpretation
`delay_jitter`	base delay τ_b (s), jitter σ_τ (s)	$\tau_b \in \{1, 3, 5, 10\}$	Variable network latency between BMS and the dispatch engine. Delayed packets arrive out of order and are consumed at their arrival time.
`spike`	magnitude μ_s (multiplier)	$\mu_s \in \{2, 3, 4, 5\}$	Transient sensor glitch producing a reading μ_s times the true value. Models ADC saturation or electromagnetic interference on current/voltage sensors.
`stale`	hold duration h (steps)	$h \in \{2, 4, 6, 8, 12\}$	Sensor freeze: the telemetry value is held constant for h consecutive steps. Models BMS firmware hang or communication bus lock-up.
`drift`	drift rate β (per step)	$\beta \in \{0.01, 0.02, 0.03, 0.05\}$	Slow, linear bias accumulating as $\xi_t = \beta t$. Models sensor calibration drift (e.g. current-sensor offset) or gradual SOH-induced SOC bias.
`reorder`	buffer size B (steps)	$B \in \{2, 4, 6, 8\}$	Out-of-order packet delivery: readings within a sliding window of B steps are randomly permuted. Models multi-path routing or priority-queue inversion on industrial networks.

c.2 Combined profile: `drift_combo`

The `drift_combo` profile activates all six fault families simultaneously at their mid-range parameter values with a configurable dropout rate d. This is the default stress profile used in the main results of the witness-domain chapter and in the bounded controller-comparison

surfaces that accompany the ORIUS publication artifacts.

Table c.2: `drift_combo` default parameter values.

Fault	**Parameter**	**Default value**
dropout	d	0.20
delay_jitter	τ_b	3 s
spike	μ_s	3×
stale	h	4 steps
drift	β	0.02 /step
reorder	B	4 steps

c.3 Replay and fault-injection implementation

Faults are injected by the CPSBench runner (`src/orius/cpsbench_iot/runner.py`) between the plant's true-state output and the controller's observation input. The injection pipeline:

1. **Plant emits** clean telemetry z_t.
2. **Fault layer** transforms $z_t \to \tilde{z}_t$ according to the active fault families and their parameters.
3. **Controller receives** degraded telemetry $\tilde{z}_t$ as its observation o_t.
4. **DC3S OQE** computes w_t from $\tilde{z}_t$ (detecting the degradation).

Fault parameters are set in `configs/dc3s.yaml` under the `fault_profile` key. Each evaluation run logs the active profile and parameter snapshot in the certificate metadata for reproducibility.

c.4 Severity mapping

Table c.3 maps each fault family to its primary impact on DC3S pipeline stages.

Table c.3: Fault-to-pipeline severity mapping.

Fault	**OQE** w_t	**RUI** m_t	**Shield IR**	**Cert. validity**
dropout	$\downarrow\downarrow$	$\uparrow\uparrow$	$\uparrow$	$\downarrow$
delay_jitter	$\downarrow$	$\uparrow$	$\uparrow$	$\downarrow$
spike	$\downarrow\downarrow$	$\uparrow\uparrow$	$\uparrow\uparrow$	$\downarrow\downarrow$
stale	$\downarrow$	$\uparrow$	$\uparrow$	$\downarrow$
drift	$\downarrow\downarrow$	$\uparrow\uparrow$	$\uparrow$	$\downarrow\downarrow$
reorder	$\downarrow$	$\uparrow$	$\uparrow$	—

$\downarrow$ = mild decrease; $\downarrow\downarrow$ = strong decrease; $\uparrow$ = mild increase; $\uparrow\uparrow$ = strong increase; — = negligible effect.

Appendix d

Artifact and Claim Index

This appendix records the canonical bridge from dissertation claims to governed repository surfaces. The prose chapters are curated, but every headline statement is expected to land on a theorem object, publication matrix, replay artifact, or explicit non-claim surface.

d.1 Claim classes and governing surfaces

Table d.1: Claim classes and their governing artifact surfaces.

Claim class	**Primary surface**	**What governs it**
Proved	Typed universal objects, assumption register, proof appendix	Theorems must point to assumptions, typed runtime objects, and the theorem-to-artifact bridge.
Validated	Publication matrices, replay tables, domain closure rows	Claims may advance only when the corresponding replay, calibration, runtime, and governance artifacts exist.
Governance	CertOS lifecycle surfaces, runtime/governance matrices	Certificate lifecycle, fallback semantics, and audit continuity remain first-class evidence, not narrative-only claims.
Roadmap	Explicit non-claims, parity blockers, promotion obligations	Future work is useful only when it stays outside headline proof/validation language until artifacts close the gap.

d.2 Canonical manuscript-facing evidence

The following generated surfaces define the defended manuscript state at any given time:

- `reports/publication/orius_equal_domain_parity_matrix.csv`
- `reports/publication/orius_submission_scorecard.csv`
- `reports/publication/orius_governance_lifecycle_matrix.csv`
- `reports/publication/orius_runtime_budget_matrix.csv`
- `reports/publication/orius_deployment_validation_scope.csv`
- `reports/publication/orius_universal_claim_matrix.csv`

d.3 Publication-facing summary

The canonical readiness summary lives in Chapter 9 because parity promotion is part of the main scientific narrative, not an appendix-only afterthought. This appendix records the governing surfaces that support that table rather than duplicating its label.

Appendix e

Safety-Case Bundle Index

This appendix maps the dissertation to a safety-case style evidence bundle without making any compliance claim. The goal is simple: a reviewer should be able to see which hazard, which argument, which evidence packet, and which non-claim boundary belong together.

Table e.1: Safety-case bundle index for the ORIUS dissertation.

Bundle	**Dissertation role**	**Primary evidence surfaces**
Hazard definition	Defines degraded observation and OASG as the governing safety problem.	Chapters 1–3, universal claim matrix, assumption register.
Theorem bundle	States what is proved under explicit assumptions.	Chapter 5, Appendix b, theorem dependency table, integrated theorem gate figure.
Benchmark bundle	Defines replay, dual trajectories, metrics, and fault families.	Chapter 6, Appendix c, domain closure matrix, parity matrix.
Witness bundle	Shows the deepest theorem-to-artifact row.	Chapters 7–8, battery tables and figures, release manifest.
Cross-domain bundle	Shows structural universality and bounded transfer.	Chapter 9, equal-domain parity matrix, submission readiness table, cross-domain support table.
Governance bundle	Shows certificate lifecycle, runtime budgets, fallback, and audit continuity.	Chapter 12, governance lifecycle matrix, runtime budget matrix, review dossier support surfaces.

Bundle	Dissertation role	Primary evidence surfaces
Boundary bundle	States what remains gated, roadmap, or non-promotional.	Chapter 13, deployment validation scope, gap matrix, exact non-claim prose.

The dissertation uses this bundle index only as an evidence-structure aid. It aligns the manuscript with disciplined safety-argument practice without implying certification or sector-specific regulatory compliance.

Extended Reading and Index

Suggested Extended Reading

Universal Safety Foundations

Angelopoulos and Bates, *Conformal Prediction: A Gentle Introduction* (2023); Vovk, Gammerman, and Shafer, *Algorithmic Learning in a Random World* (2005); Shafer and Vovk, "A Tutorial on Conformal Prediction" (JMLR 2008).

Conformal Prediction and Calibration

Romano, Patterson, and Candès, "Conformalized Quantile Regression" (NeurIPS 2019); Gibbs and Candès, "Adaptive Conformal Inference Under Distribution Shift" (NeurIPS 2021); Barber et al., "Conformal Prediction Beyond Exchangeability" (AoS 2023); Tibshirani et al., "Conformal Prediction Under Covariate Shift" (2019); Zaffran et al., "Adaptive Conformal Predictions for Time Series" (2022).

Runtime Assurance and Safe Control

Ames et al., "Control Barrier Functions: Theory and Applications" (2019); Bloem et al., "Shield Synthesis" (TACAS 2015); Wabersich and Zeilinger, "A Predictive Safety Filter for Learning-Based Control" (2021); Alshiekh et al., "Safe Reinforcement Learning via Shielding" (2018); Desai et al., "SOTER" (IROS 2019).

Degraded Observation and CPS Trust

Chandola, Banerjee, and Kumar, "Anomaly Detection: A Survey" (2009); Liu, Ting, and Zhou, "Isolation Forest" (2008); Teixeira et al., "A Secure Control Framework for Resource-Limited Adversaries" (2015); Cardenas et al., "Attacks Against Process Control Systems" (2011); Giraldo et al., "A Survey of Physics-Based Attack Detection in Cyber-Physical Systems" (2018).

Domain Instantiations

Energy and storage: Tarascon and Armand (2001), Dunn et al. (2011), Rosewater et al. (2019), Hossain et al. (2020). Automated driving and robotics: Koopman and

Wagner (2016), Shalev-Shwartz et al. (2017), Althoff and Dolan (2014), Herbert et al. (2017). Industrial systems: Qin and Badgwell (2003), Yin et al. (2014), Isermann (2005), Zhang and Jiang (2008). Healthcare CPS: Lee and Sokolsky (2010), Saeed et al. (2011), Clifford et al. (2009), FDA SaMD guidance (2019). Navigation and planning: Kaelbling, Littman, and Cassandra (1998), Julier and Uhlmann (2004), Rockafellar and Uryasev (2000). Aerospace assurance: RTCA DO-178C (2011), Rushby (1993), Habli and Kelly (2010), Srinivasan and Sanner (2020).

Reproducibility and Scientific Software

Gneiting and Raftery (2007); Makridakis et al. (2020); Lei et al. (2018); Barber, Candès, Ramdas, and Tibshirani (2023). Use the locked artifact index and publication manifests for all thesis-grade claims.

Key Symbol Index

The dissertation keeps its canonical notation in the core theory and assumption surfaces: see Chapters 3–5 and Appendix a.

Artifact Index

The artifact, figure, and table index is maintained in the repository at `reports/publication/` alongside the locked evaluation outputs.

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